

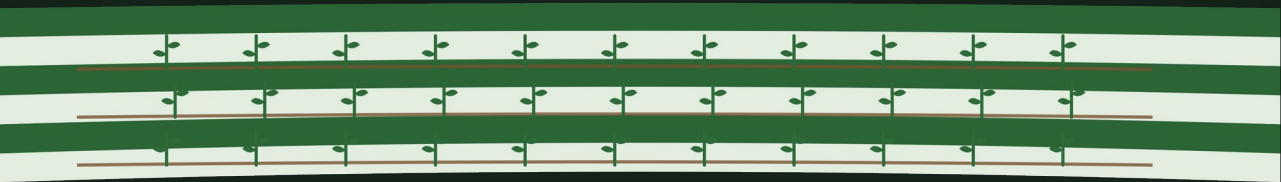
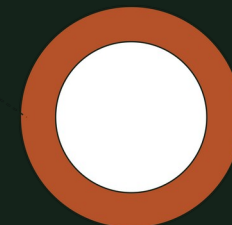
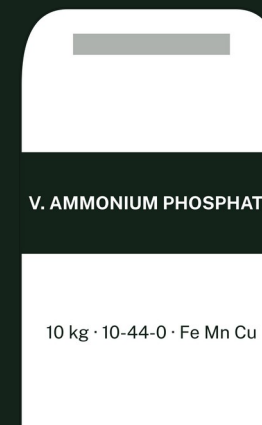


DEALER TRAINING · SEPTEMBER 2026

V. Ammonium Phosphate

10-44-0 with iron, manganese and copper, fully soluble, 10 kg, the mid-life phosphate through the water.

For VAN dealers and the field team, written so a dealer can repeat it to a grower. Lahore.



What a dealer says in 30 seconds.

V. Ammonium Phosphate is 10-44-0, fully soluble, in a 10 kg bag, with iron, manganese and copper carried with the phosphate. It is the mid-life phosphate. The basal bag went in at land preparation, months before the crop asks hardest for phosphorus; on calcareous soil above pH 8 the phosphate that dissolved from it meets calcium within centimetres. This bag puts a second, soluble dose through the water at early growth or grand growth, when the crop is building tillers, bolls or tubers. The nitrogen is ammonium, the form the root takes up as it is. Dissolve it fully, 1 part to 10 parts water, and release it where the water enters the field so it travels the whole acre. 1 bag an acre on most crops; potato and onion take it twice. It is registered as V-Phosphate 10-44-0; older bags in the market may still read 12-44-0.

The 4 things it must contain

1

The bag: 10-44-0, 10 kg, fully soluble, PS 933-2026. 4.4 kg of P_2O_5 and 1 kg of nitrogen in every bag.

2

The job: the mid-life phosphate, through the water at early growth or grand growth. The basal bag is a different bag at a different stage.

3

The method: dissolve fully, 1 to 10 in water, release where the water enters the field. Every programme row says fertigation.

4

The name: registered as V-Phosphate 10-44-0, sold as V. Ammonium Phosphate. Same product. The registration governs.

Say it in the grower's own words. Every figure in these 30 seconds is on the bag, in the published crop programme, or in the named evidence on the slides that follow.

THE BAG

What the bag says, and what it is registered as.

10 kg

the bag. 4.4 kg of P₂O₅ and 1.0 kg of nitrogen in every bag, from the declared analysis.

P₂O₅ 44%

phosphate, declared as NPK 10-44-0, fully soluble.

N 10%

nitrogen, as ammonium: the form the root takes up as it is, with no conversion step.

Fe • Mn • Cu

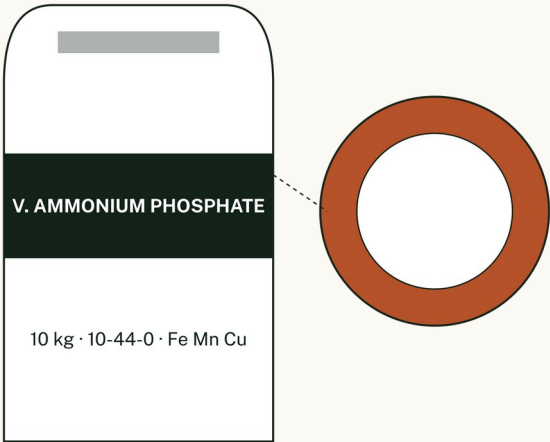
iron, manganese and copper carried with the phosphate, as van.com.pk names them. Named, never numbered: no figure is on the bag.

Mid-life

the stage: early growth or grand growth, through the water. In 27 of 28 published programmes, always by fertigation.

LAB 336

every batch is released through VAN Lab, PNAC accredited, before it ships.



Registered as	Pakistan Standard	Manufacturing licence
V. Ammonium Phosphate	PS 933-2026	CM/L-4297/2026

No expiry date is printed in this deck. The licence number is what a dealer quotes.

A GROWER'S QUESTION

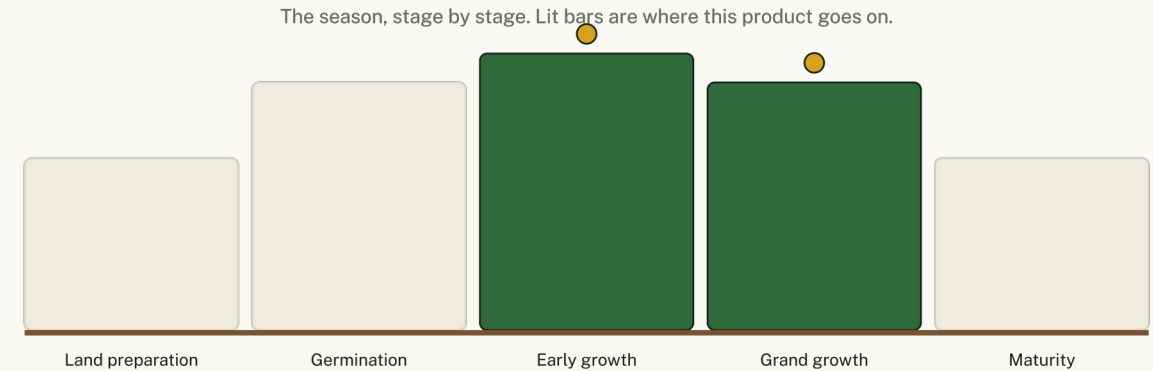
“I put my phosphate in at sowing. Why does the programme ask me to buy more in the middle of the season?”

The grower did the right thing at land preparation and reads the programme asking for a second phosphate bag at early or grand growth. He wants to know why the first bag was not enough, why this one goes through the water instead of the ground, and what the iron is doing in it. The answer is 3 ideas: the crop asks twice, soluble phosphate through the water reaches the whole acre, and the micronutrients ride with it.

The 3 answers on the next 3 slides, 1 idea each.

The crop asks for phosphorus twice. The basal bag answers once.

Phosphorus builds the root in the first weeks, and the basal bag is placed for that. Then the crop asks again, harder, when it builds tillers, bolls, tubers or fruit. By then the basal granule has been in the soil for months, and on calcareous soil above pH 8 the phosphate that dissolved out of it met calcium within a few centimetres and dropped out of solution. The second demand is met by a second dose, placed when the crop asks for it. Every programme that names V. Ammonium Phosphate puts it at early growth or grand growth, never at land preparation.



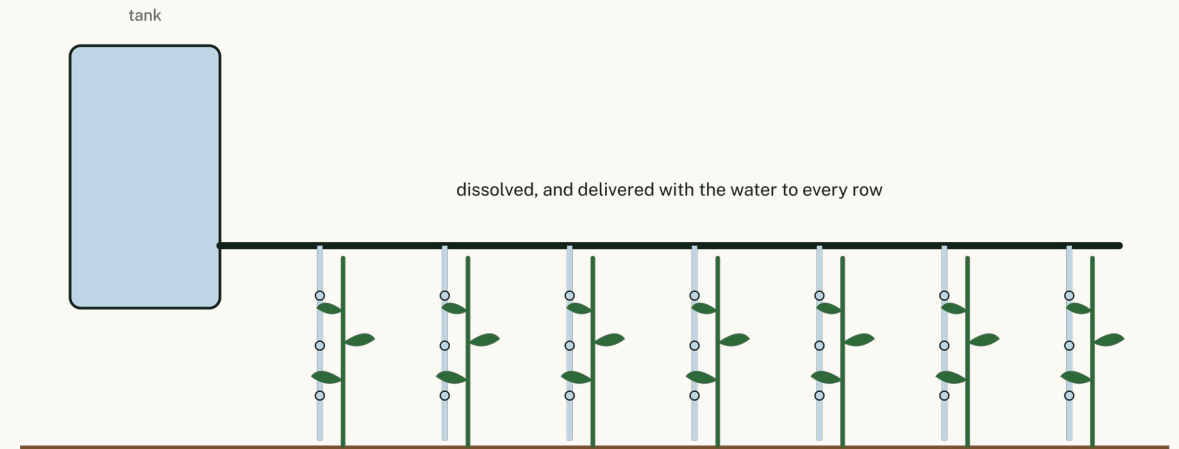
The stage names are the programme's. Which 1 or 2 stages carry this bag for your crop is on the programme slides.

Say it like this: "Your sowing bag built the root. This bag feeds the crop when it asks again, in the middle. 2 jobs, 2 bags."

Fully soluble, through the water: 1 bag reaches the whole acre.

The bag dissolves completely. Mixed 1 part to 10 parts water, 10 kg in 100 litres, and released where the irrigation water enters the field, it travels with the water across the whole acre and into the wet root zone, instead of the strip a spreader covered. The solution is slightly acidic, pH 5 to 6.5 from VAN's product data, so it does not push already-alkaline irrigation water further up. The nitrogen is ammonium, taken up as it is, and as a root takes up ammonium it acidifies the soil beside itself a little, which helps the phosphate next to it stay in solution.

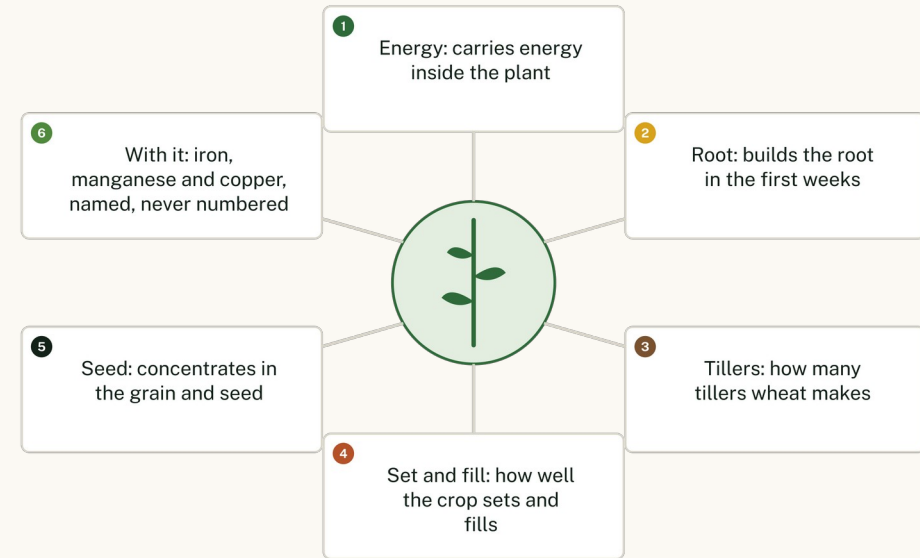
Say it like this: "Dissolve the whole bag, 1 to 10 in water, and let it in where the water enters the field. The water carries it to every root."



The drum at the head of the watercourse. No figure for how much of the dose reaches the root is printed, because none is measured.

Iron, manganese and copper ride with the phosphate. Named, never numbered.

Phosphorus and the micronutrients pull against each other: a heavy dose of phosphate on a soil already low in zinc or iron makes that shortage worse, and calcareous soil locks iron and manganese the same way it locks phosphate. So the phosphate carries its own iron, manganese and copper with it, as van.com.pk names them, and they land where and when the phosphate lands. No figure is printed for them, because none is on the bag. The crop still gets its zinc from the zinc product the programme names; this bag does not replace it.



The 6 jobs of phosphorus from the 17-nutrients page on van.com.pk, and the 3 micronutrients as passengers. A picture of the idea, not a measurement.

Say it like this: “The iron and the manganese come in the same bag as the phosphate, so the phosphate does not leave the crop short of them. No percentage; they are named, not numbered.”

What is verified for V. Ammonium Phosphate, and what is not.

No trial of the product is on VAN's record. What a dealer may state as fact is the registration, the bag, the soil record and the published soil chemistry, each as what it is.

Item	What it says	How a dealer may use it
The registration	Registered as V-Phosphate 10-44-0 against Pakistan Standard PS 933-2026, manufacturing licence CM/L-4297/2026. Sold as V. Ammonium Phosphate.	As fact. The registration governs the analysis.
The bag	NPK 10-44-0, 10 kg, fully soluble powder. Iron, manganese and copper named on van.com.pk, not numbered. Bags printed before the re-registration may still read 12-44-0.	As fact. On an old bag: same product, the certificate governs.
The soil record (Punjab, 770,160 samples)	69.1% of samples above pH 8; 97.9% below adequate in available phosphorus; 59.0% low in zinc. Iron and manganese are on the soil report and the Soil Atlas.	As fact about the soil. Never as product performance.
The soil chemistry (published)	Above pH 8 with free lime, dissolved phosphate meets calcium within centimetres and drops out of solution; the first calcium phosphates are still fairly available, the loss is their ageing into rock-like forms.	As general soil chemistry. Not as a percentage fixed.
The programmes	27 of 28 published programmes name it, at early growth, grand growth or maturity, every row by fertigation, 10 kg bag.	As fact. The stage and rate are the programme's.
Not on file	A trial; a yield figure; a micronutrient percentage; a tank-mix compatibility chart; a sourced national phosphorus-deficiency percentage beyond the Punjab survey.	Not said.

Sources: the PSQCA register as van.com.pk prints it; the bag and the product page on van.com.pk; the Punjab soil testing programme (36 district workbooks, 770,160 samples); Tunesi 1999 and Freeman and Rowell 1981 for the calcium chemistry; the crop programmes on van.com.pk.

The bag, in the grower’s own back-of-envelope.

Price from the VAN farmer price sheet, Aug 2026, delivered farmer price. P₂O₅ and nitrogen from the declared analysis. The micronutrients carry no figure.

V. Ammonium Phosphate, 10-44-0	Weight	P ₂ O ₅	Nitrogen	Rupees
1 bag	10 kg	4.4 kg	1.0 kg	7,100
½ bag	5 kg	2.2 kg	0.5 kg	3,550
Potato, 1 bag at early growth + ½ at grand growth	15 kg	6.6 kg	1.5 kg	10,650

On wheat the grower spends 7,100 rupees an acre to put 4.4 kg of soluble P₂O₅ through the water at early growth. On potato it is 10,650 across 2 stages. That is the counter arithmetic. What the crop does with it is not a number VAN prints.

What the programme puts through the water, per acre: the field crops.

Bags per acre from the published crop programmes; P₂O₅ from the declared analysis; rupees at the VAN farmer price sheet, Aug 2026. Every row is by fertigation.

Crop	Bags per acre	P ₂ O ₅ put in	Rupees per acre	Stage, as printed
Wheat	1 bag of 10 kg	4.4 kg	7,100	Early growth
Maize	1 bag of 10 kg	4.4 kg	7,100	Grand growth
Basmati and hybrid rice	1 bag of 10 kg	4.4 kg	7,100	Early growth
Cotton	1 bag of 10 kg	4.4 kg	7,100	Grand growth
Sugarcane	1 bag of 10 kg	4.4 kg	7,100	Early growth
Sugarcane ratoon	1 bag of 10 kg	4.4 kg	7,100	Grand growth

That is the grower's mid-life phosphate line for the field crops, in his own arithmetic. It is added to the basal bag at land preparation, never swapped for it.

The crop programmes on van.com.pk, V. Ammonium Phosphate (V-Phosphate) rows; 10 kg bag at PKR 7,100. Rupees = bags × 7,100.

What the programme puts through the water, per acre: the vegetables and the orchard.

Bags per acre from the published crop programmes; P₂O₅ from the declared analysis; rupees at the VAN farmer price sheet, Aug 2026. Potato and onion take it twice.

Crop	Bags per acre	P ₂ O ₅ put in	Rupees per acre	Stage, as printed
Potato	1 + ½ bag	6.6 kg	10,650	Early growth, then grand growth
Onion	1 + 1 bag	8.8 kg	14,200	Early growth (20 days after transplanting), grand growth (40 days)
Tomato	1 bag of 10 kg	4.4 kg	7,100	Grand growth
Chili	1 bag of 10 kg	4.4 kg	7,100	Early growth
Citrus	1 bag of 10 kg	4.4 kg	7,100	Grand growth

Garlic and the other 15 programmes that name it are on van.com.pk with the same pattern: early growth or grand growth, through the water.

The crop programmes on van.com.pk, V. Ammonium Phosphate (V-Phosphate) rows; 10 kg bag at PKR 7,100. Rupees = bags × 7,100.

The phosphate account for an acre of wheat: 2 bags in, 1 harvest out.

What 40 maunds of wheat (1.6 tonnes) takes off the field, from the IPNI removal tables the site’s balance block uses, beside the basal and mid-life bags the programme puts on.

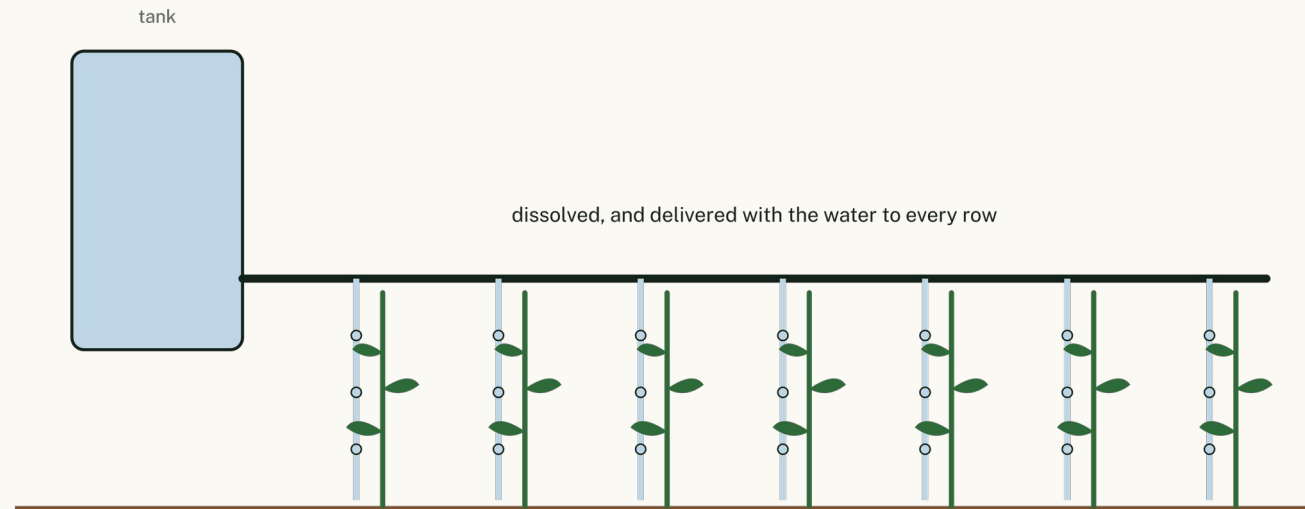
Wheat, 1 acre at 40 maunds	P ₂ O ₅ per tonne of grain	P ₂ O ₅ at 1.6 t	Nitrogen at 1.6 t
Grain taken off	8.0 kg	12.8 kg	30 kg
Straw (bhusa) taken off too, per tonne of grain	2.7 kg	4.3 kg	19 kg
Grain and straw together	10.7 kg	17.1 kg	50 kg
Green Phosphate, 1 bag at land preparation, puts in		16 kg	3 kg
V. Ammonium Phosphate, 1 bag at early growth, puts in		4.4 kg	1 kg

The 2 bags together put in 20.4 kg of P₂O₅ against 17.1 kg the harvest takes off with the bhusa. The kilograms balance on paper; on calcareous soil what the root meets of the basal 16 kg is the mechanism slide, and the soluble 4.4 kg at early growth is placed when the crop asks for it.

IPNI nutrient removal tables (metric, 01/15): wheat grain P₂O₅ 8.0 and N 19; straw P₂O₅ 2.7 and N 12 per tonne of grain. 40 maunds × 40 kg = 1.6 t. Bags: the wheat programme on van.com.pk. 16 kg = 50 × 0.32; 4.4 kg = 10 × 0.44.

HOW IT GOES ON

Through the water, at early growth or grand growth. Dissolve it fully first.



Weigh it, then dissolve it

1 bag an acre on most crops; weigh it, a guessed dose is the commonest error in the field. Into the drum or tank at 1 part product to 10 parts water: 10 kg in 100 litres. Stir until nothing is left on the bottom. It dissolves completely; if anything is left, it has not been given long enough.

Release where the water enters the field

Open the tap so the drum empties across the whole irrigation, not in 1 slug, with the water already flowing. On drip, into the tank the same way. The water carries it to every root on the acre.

At the programme's stage, never in place of the basal bag

Early growth for wheat, rice, sugarcane and chili; grand growth for maize, cotton, tomato and citrus; potato and onion twice. Every row says fertigation. It is not a basal granule and not a foliar; no foliar rate is printed.

OBJECTION 1 OF 8

“7,100 for 10 kg? A bag of DAP is 50 kg.”

The dealer says, out loud

Write both on your paper and then write the job beside each. DAP is a basal granule; it goes in the ground at sowing. This bag is 4.4 kg of soluble phosphate that goes through the water in the middle of the season, when the crop asks for phosphorus again and the sowing bag has been in a calcareous soil for months. They are 2 stages, and the programme uses a basal bag and this one. Do not compare the kilograms; compare when and how each reaches the crop.

The fact behind it

- V. Ammonium Phosphate 10 kg PKR 7,100 (VAN farmer price sheet, Aug 2026). DAP: the counter price, not on the VAN sheet.
- 4.4 kg P_2O_5 and 1 kg N per bag (the declared analysis). Wheat: 1 bag at early growth (the crop programmes on van.com.pk).
- The 4 phosphate routes on van.com.pk: this is the fertigation route at mid-life; the basal routes are Green Phosphate and Fusion Phosphate.

Never say: “It replaces your DAP” or a percentage of phosphate saved.

“I already gave phosphate at sowing. Why again?”

The dealer says, out loud

Because the crop asks twice. The first weeks build the root, and the sowing bag was placed for that. Then the crop builds tillers, bolls or tubers and asks harder, and by then the basal granule has sat in a soil above pH 8 for months, where the phosphate that dissolved met calcium within centimetres. The programme meets the second demand with a soluble dose through the water at early growth or grand growth. It is added to the basal bag, not swapped for it.

The fact behind it

- Phosphorus builds the root first, then decides tillers, set and fill (the 17 nutrients page on van.com.pk).
- Every V. Ammonium Phosphate row is at early growth, grand growth or maturity; Green Phosphate is the land-preparation row on the same programmes (van.com.pk).
- Above pH 8 with free lime, dissolved phosphate meets calcium within centimetres (Tunesi 1999; Freeman and Rowell 1981).

Never say: “80% of your sowing bag was lost.” No percentage is printed.

“Can I just broadcast it dry, or spray it on the leaf?”

The dealer says, out loud

Every programme row says fertigation, and that is how it is built: fully soluble, mixed 1 to 10, released with the water. Spread dry it is a small bag on a large field and the calcium reaches it before the root does. Foliar is not a method any programme names for it, and I will not give you a foliar rate. If you cannot fertigate, the basal phosphate at land preparation is the bag for your field.

The fact behind it

- The product page lists 1 method, fertigation; all 33 programme rows say fertigation (van.com.pk).
- It goes in at the drum, not at the bag: dissolve, stir, meter it in with the water flowing.
- No foliar rate is published for it anywhere on van.com.pk.

Never say: A foliar rate, or “broadcast it if you have no water”.

OBJECTION 4 OF 8

“Can I mix it in the tank with urea, or with my zinc?”

The dealer says, out loud

VAN publishes no tank-mix chart for this product, so I will not guess one for you. Dissolve it on its own, 1 to 10 in water, and release it with the irrigation. If you want to put 2 products through the same water, ask VAN before you do it, with both bags in hand.

The fact behind it

- No compatibility chart for this product is published on van.com.pk.
- The published method: dissolve, agitate, meter it in, on its own.
- Tank-mix and compatibility charts are among the documents VAN shares only on request.

Never say: “It mixes with anything.”

“My bag says V-Ammonium Phosphate 12-44-0. Is this the same thing?”

The dealer says, out loud

Yes. The product was re-registered as V-Phosphate 10-44-0 under licence CM/L-4297/2026, and it is sold as V. Ammonium Phosphate. Bags printed before that still read 12-44-0. Same product, same job in the programme. The registration and the batch certificate govern the analysis: read the batch number off your bag and VAN sends you that batch's certificate, free.

The fact behind it

- Registered as V-Phosphate 10-44-0, marketed as V. Ammonium Phosphate; PS 933-2026, CM/L-4297/2026 (the regulatory table on van.com.pk).
- Artwork printed before the re-registration reads 12-44-0.
- The certificate of analysis for any batch is free on request, by batch number.

Never say: “The old bag has more nitrogen, buy that one.”

“My leaves are yellow between the veins. Will the iron in this fix it?”

The dealer says, out loud

The iron, manganese and copper in this bag ride with the phosphate so that the phosphate does not leave the crop short of them. They are named, not numbered, and VAN sells no cure in this bag. Yellow between the veins on the new leaf on calcareous ground is often an iron story, and the programme has a micronutrient product for it. Put this bag on at its stage for its job; ask VAN about the yellowing with a photo of the leaf.

The fact behind it

- Iron shortage shows as young leaves yellow between the veins with the veins sharply green; high pH locks iron (the 17 nutrients page on van.com.pk).
- Iron, manganese and copper are named on the product page, with no figure (van.com.pk).
- VAN's micronutrient products for iron are named on the same page.

Never say: “It greens up the crop in a week” or a micronutrient percentage.

“How many bags an acre, and when?”

The dealer says, out loud

What the programme says: 1 bag on most crops, at early growth for wheat, rice, sugarcane and chili, at grand growth for maize, cotton, tomato and citrus. Potato takes 1 bag at early growth and $\frac{1}{2}$ at grand growth. Onion takes 1 bag at 20 days after transplanting and 1 at 40. Through the water, every time. More bags than the programme is more phosphate than the crop asked for at that stage.

The fact behind it

- Rates per acre as published for 12 crops on the programme slides (the crop programmes on van.com.pk).
- All rows by fertigation, 10 kg bag.
- Potato 1 + $\frac{1}{2}$, onion 1 + 1, garlic $\frac{1}{2}$ + 1 + 1; every other crop 1 bag once.

Never say: “Double it if the crop looks weak.”

“Is it registered? What is the licence?”

The dealer says, out loud

Pakistan Standard PS 933-2026 for multnutrient fertilizer, manufacturing licence CM/L-4297/2026, registered as V-Phosphate 10-44-0 with VAN as the maker. Every batch is tested at VAN Lab, PNAC LAB 336, before it ships. Read the batch number off the bag and VAN sends you that batch's certificate, free.

The fact behind it

- V-Phosphate: PS 933-2026, CM/L-4297/2026 (the regulatory table on van.com.pk).
- Every batch is released through VAN Lab, PNAC LAB 336 (the corporate profile).
- The certificate of analysis for any batch is free on request, by batch number.

Never say: An expiry date, or the micronutrients as certificate lines unless the certificate carries them.

V. Ammonium Phosphate in the published programmes, per acre.

Crop	Stage	Rate per acre	Method	Pack
Wheat	Early Growth	1 bag	Fertigation	10 kg
Maize (Corn)	Grand Growth	1 bag	Fertigation	10 kg
Basmati Rice	Early Growth	1 bag	Fertigation	10 kg
Hybrid Rice	Early Growth	1 bag	Fertigation	10 kg
Cotton	Grand Growth	1 bag	Fertigation	10 kg
Sugarcane, February planting	Early Growth	1 bag	Fertigation	10 kg
Ratoon Sugarcane	Grand Growth	1 bag	Fertigation	10 kg

Rows as published in van.com.pk's crop programmes, 12 of the 27 programmes that name V. Ammonium Phosphate. Every row is by fertigation at early growth, grand growth or maturity; the programme prints the product as "V. Ammonium Phosphate (V-Phosphate)". A rate is per acre; the crop page scales it to the field.

V. Ammonium Phosphate in the published programmes, continued.

Crop	Stage	Rate per acre	Method	Pack
Potato	Early Growth	1 bag	Fertigation	10 kg
Potato	Grand Growth	½ bag	Fertigation	10 kg
Tomato	Grand Growth	1 bag	Fertigation	10 kg
Chili	Early Growth	1 bag	Fertigation	10 kg
Onion	Early Growth (20 DOT)	1 bag	Fertigation	10 kg
Onion	Grand Growth (40 DOT)	1 bag	Fertigation	10 kg
Citrus, 5 years and over	Grand Growth	1 bag	Fertigation	10 kg

Rows as published in van.com.pk's crop programmes, 12 of the 27 programmes that name V. Ammonium Phosphate. Every row is by fertigation at early growth, grand growth or maturity; the programme prints the product as "V. Ammonium Phosphate (V-Phosphate)". A rate is per acre; the crop page scales it to the field.

WHAT VAN WILL NOT CLAIM FOR IT

A dealer who says any of these is speaking for himself, not for VAN.

1

A yield figure, or that it beats urea phosphate, DAP or any other phosphate. No trial of the product exists.

2

That 80% of the sowing phosphate is fixed, or a national phosphorus-deficiency percentage beyond the Punjab survey's 97.9% below adequate.

3

A percentage for iron, manganese or copper. They are named, never numbered.

4

A foliar rate or a dry broadcast use. Every programme row is fertigation.

5

A tank-mix compatibility claim. No chart is published.

6

That it replaces the basal phosphate, or cures iron chlorosis. It is the mid-life phosphate, added to the basal bag.

Every batch is tested. The bag carries the proof.

1 · The batch number

It is printed on the bag. Read it to the grower and show him where it is. Every bag of every batch carries one.

2 · The certificate of analysis

For any batch, free on request. Send the batch number off the bag to kisan@van.com.pk or the helpline, and the certificate for that batch comes back. No sample to post.

3 · The laboratory

VAN Lab, PNAC LAB 336, accredited to ISO/IEC 17025:2017. Every batch is released through it before it ships. The certificate carries the laboratory's number.

What a certificate for V. Ammonium Phosphate shows

The declared analysis tested for the batch: P_2O_5 and nitrogen against 10-44-0, with the batch number, the Pakistan Standard PS 933-2026 and licence CM/L-4297/2026. The registered name on the certificate is V-Phosphate 10-44-0; the bag in the market may read V. Ammonium Phosphate, and older artwork 12-44-0. The certificate governs. No certificate for a batch was on file for this deck, so no batch reading is printed here; the grower asks for his own batch's certificate.

The lines a dealer says in Urdu.

The first line is the promise word from the VAN brand master. The other 3 are the deck's key sentences, translated in a first pass and not yet reviewed by VAN. Say them, and send corrections to partner@van.com.pk.

REACH. The promise word for phosphorus.

Urdu first pass, unreviewed

پہلے جر، پھر سب کچھ

The crop asks for phosphorus again in the middle of the season.

Urdu first pass, unreviewed

فصل درمیانی موسم میں دوبارہ فاسفورس مانگتی ہے

Dissolve it fully, 1 to 10 in water, and let it in while the water is flowing.

Urdu first pass, unreviewed

پورا گھول لیں، 1 حصہ کھاد 10 حصے پانی، اور چلتے پانی کے ساتھ دیں

Iron, manganese and copper travel to the root with the phosphorus.

Urdu first pass, unreviewed

لوہا، منگنیز اور تانبا فاسفورس کے ساتھ جر تک جاتے ہیں

Write to the address for what you need.

If you grow

kisan@van.com.pk

A crop plan, product rates, a certificate for your batch, a dealer near you.

If you sell

partner@van.com.pk

Dealers, distributors and the trade. Stock, training, corrections to this deck.

Everything else

info@van.com.pk

Supply, the laboratory, a plant visit.

Helpline 0300 5003041 · www.van.com.pk

Vital Agri Nutrients (Pvt) Ltd, Lahore. 1 helpline, 3 addresses. No group broadcasts: a grower who writes gets 1 reply.

V. Ammonium Phosphate: the mid-life phosphate, dissolved and put through the water at early or grand growth. Added to the basal bag, never in place of it.

